

Level 1 Training 16: Contest #1

A. Domino piling

time limit per test: 2 seconds

memory limit per test: 256 megabytes

input: standard input

output: standard output

You are given a rectangular board of $M \times N$ squares. Also you are given an unlimited number of standard domino pieces of 2×1 squares. You are allowed to rotate the pieces. You are asked to place as many dominoes as possible on the board so as to meet the following conditions:

1. Each domino completely covers two squares.
2. No two dominoes overlap.
3. Each domino lies entirely inside the board. It is allowed to touch the edges of the board.

Find the maximum number of dominoes, which can be placed under these restrictions.

Input

In a single line you are given two integers M and N — board sizes in squares ($1 \leq M \leq N \leq 16$).

Output

Output one number — the maximal number of dominoes, which can be placed.

Sample test(s)

input
2 4
output
4

input
3 3
output
4

B. Drazil and Date

time limit per test: 1 second

memory limit per test: 256 megabytes

input: standard input

output: standard output

Someday, Drazil wanted to go on date with Varda. Drazil and Varda live on Cartesian plane. Drazil's home is located in point $(0, 0)$ and Varda's home is located in point (a, b) . In each step, he can move in a unit distance in horizontal or vertical direction. In other words, from position (x, y) he can go to positions $(x + 1, y)$, $(x - 1, y)$, $(x, y + 1)$ or $(x, y - 1)$.

Unfortunately, Drazil doesn't have sense of direction. So he randomly chooses the direction he will go to in each step. He may accidentally return back to his house during his travel. Drazil may even not notice that he has arrived to (a, b) and continue travelling.

Luckily, Drazil arrived to the position (a, b) successfully. Drazil said to Varda: "It took me exactly s steps to travel from my house to yours". But Varda is confused about his words, she is not sure that it is possible to get from $(0, 0)$ to (a, b) in exactly s steps. Can you find out if it is possible for Varda?

Input

You are given three integers a , b , and s ($-10^9 \leq a, b \leq 10^9$, $1 \leq s \leq 2 \cdot 10^9$) in a single line.

Output

If you think Drazil made a mistake and it is impossible to take exactly s steps and get from his home to Varda's home, print "No" (without quotes).

Otherwise, print "Yes".

Sample test(s)

input
5 5 11
output
No

input
10 15 25
output
Yes

input
0 5 1
output
No

input
0 0 2
output
Yes

Note

In fourth sample case one possible route is: $(0, 0) \longrightarrow (0, 1) \longrightarrow (0, 0)$.

C. Bit++

time limit per test: 1 second

memory limit per test: 256 megabytes

input: standard input

output: standard output

The classic programming language of Bitland is Bit++. This language is so peculiar and complicated.

The language is that peculiar as it has exactly one variable, called x . Also, there are two operations:

- Operation `++` increases the value of variable x by 1.
- Operation `--` decreases the value of variable x by 1.

A statement in language Bit++ is a sequence, consisting of exactly one operation and one variable x . The statement is written without spaces, that is, it can only contain characters `"+"`, `"-"`, `"X"`. Executing a statement means applying the operation it contains.

A programme in Bit++ is a sequence of statements, each of them needs to be executed. Executing a programme means executing all the statements it contains.

You're given a programme in language Bit++. The initial value of x is 0. Execute the programme and find its final value (the value of the variable when this programme is executed).

Input

The first line contains a single integer n ($1 \leq n \leq 150$) — the number of statements in the programme.

Next n lines contain a statement each. Each statement contains exactly one operation (`++` or `--`) and exactly one variable x (denoted as letter `«X»`). Thus, there are no empty statements. The operation and the variable can be written in any order.

Output

Print a single integer — the final value of x .

Sample test(s)

input
1 ++X
output
1

input
2 X++ --X
output
0

D. Tricky Sum

time limit per test: 1 second

memory limit per test: 256 megabytes

input: standard input

output: standard output

In this problem you are to calculate the sum of all integers from 1 to n , but you should take all powers of two with minus in the sum.

For example, for $n = 4$ the sum is equal to $-1 - 2 + 3 - 4 = -4$, because 1, 2 and 4 are 2^0 , 2^1 and 2^2 respectively.

Calculate the answer for t values of n .

Input

The first line of the input contains a single integer t ($1 \leq t \leq 100$) — the number of values of n to be processed.

Each of next t lines contains a single integer n ($1 \leq n \leq 10^9$).

Output

Print the requested sum for each of t integers n given in the input.

Sample test(s)

input
2 4 1000000000
output
-4 4999999998352516354

Note

The answer for the first sample is explained in the statement.

E. Duff and Meat

time limit per test: 1 second

memory limit per test: 256 megabytes

input: standard input

output: standard output

Duff is addicted to meat! Malek wants to keep her happy for n days. In order to be happy in i -th day, she needs to eat exactly a_i kilograms of meat.



There is a big shop uptown and Malek wants to buy meat for her from there. In i -th day, they sell meat for p_i dollars per kilogram. Malek knows all numbers $a_1, \dots, a_n$ and $p_1, \dots, p_n$. In each day, he can buy arbitrary amount of meat, also he can keep some meat he has for the future.

Malek is a little tired from cooking meat, so he asked for your help. Help him to minimize the total money he spends to keep Duff happy for n days.

Input

The first line of input contains integer n ($1 \leq n \leq 10^5$), the number of days.

In the next n lines, i -th line contains two integers a_i and p_i ($1 \leq a_i, p_i \leq 100$), the amount of meat Duff needs and the cost of meat in that day.

Output

Print the minimum money needed to keep Duff happy for n days, in one line.

Sample test(s)

input
3 1 3 2 2 3 1
output
10

input
3 1 3 2 1 3 2
output
8

Note

In the first sample case: An optimal way would be to buy 1 kg on the first day, 2 kg on the second day and 3 kg on the third day.

In the second sample case: An optimal way would be to buy 1 kg on the first day and 5 kg (needed meat for the second and third day) on the second day.

F. Watermelon

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

One hot summer day Pete and his friend Billy decided to buy a watermelon. They chose the biggest and the ripest one, in their opinion. After that the watermelon was weighed, and the scales showed w kilos. They rushed home, dying of thirst, and decided to divide the berry, however they faced a hard problem.

Pete and Billy are great fans of even numbers, that's why they want to divide the watermelon in such a way that each of the two parts weighs even number of kilos, at the same time it is not obligatory that the parts are equal. The boys are extremely tired and want to start their meal as soon as possible, that's why you should help them and find out, if they can divide the watermelon in the way they want. For sure, each of them should get a part of positive weight.

Input

The first (and the only) input line contains integer number w ($1 \leq w \leq 100$) — the weight of the watermelon bought by the boys.

Output

Print YES, if the boys can divide the watermelon into two parts, each of them weighing even number of kilos; and NO in the opposite case.

Sample test(s)

input
8
output
YES

Note

For example, the boys can divide the watermelon into two parts of 2 and 6 kilos respectively (another variant — two parts of 4 and 4 kilos).